Other AWS Compute Services Self-guided

Skills you will learn

Cloud Computing: Compute & Networking

Introduction

AWS provides a diverse range of compute services beyond traditional virtual machines (EC2). This lesson explores additional compute solutions, including container-based and serverless computing, to help you understand the different options available for running applications on AWS.

Learning Objectives

By the end of this lesson, you will be able to:

1. **Identify different AWS compute services** beyond EC2.
2. **Understand the functionalities and benefits** of these compute services.
3. **Explore the advantages of serverless computing.**
4. **Discover AWS's portfolio of containerized and serverless compute services.**

Compute Services in AWS

AWS offers multiple compute services to suit different application needs. These services can be categorized into:

- **Container-based compute solutions**
- **Serverless compute solutions**

1. Amazon Elastic Container Service (ECS)

Amazon ECS is a **fully managed container orchestration service** that enables you to run and manage Docker containers. It allows for easy deployment and scaling of containerized applications across AWS infrastructure.

Key Features:

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- Supports **Docker containers**
- Runs on **EC2 instances** or **AWS Fargate**
- Integrates with **IAM, CloudWatch, and ALB**

2. Amazon Elastic Kubernetes Service (EKS)

Amazon EKS is a **managed Kubernetes service** that simplifies the deployment and management of Kubernetes clusters on AWS.

Key Features:

- Provides **fully managed control planes** for Kubernetes clusters
- Integrates with AWS services like **IAM, VPC, and Load Balancers**
- Supports **high availability and automatic scaling**

3. AWS Fargate - Serverless Containers

AWS Fargate is a **serverless compute engine for containers** that allows users to run containers without managing the underlying infrastructure.

Key Features:

- Eliminates the need to provision and manage EC2 instances
- Scales automatically based on workload demand
- Supports both **ECS and EKS**

4. Serverless Compute

Serverless computing allows users to run applications without provisioning or managing servers. AWS provides several **serverless services** to help developers focus on writing code instead of managing infrastructure.

Benefits of Serverless Compute:

- **Consumption-based pricing** - You only pay for what you use.
- **Automatic scaling** - Services scale based on demand.
- **No infrastructure management** - AWS handles the infrastructure.

5. AWS Lambda - Event-Driven Compute

AWS Lambda is a serverless compute service that allows you to run code in response to events without provisioning or managing servers.

Key Features:

- Supports multiple languages: **Python, Node.js, Java, Go, C#, and Ruby**
- Automatically scales based on incoming requests

- Charges only for execution time (pay-per-use)

6. AWS Step Functions - Serverless Workflow Automation

AWS Step Functions is a **serverless orchestration service** that lets you coordinate AWS services into workflows.

Common Use Cases:

- **Automating workflows** between AWS services
- **Orchestrating microservices**
- **Building data processing pipelines**

7. Amazon Simple Queue Service (SQS) - Managed Message Queue

Amazon SQS is a **fully managed message queuing service** that allows decoupling of microservices and distributed applications.

Key Features:

- Stores and processes messages reliably
- Supports **Standard and FIFO queues**
- Fully integrates with **Lambda, ECS, and Step Functions**

8. Amazon Simple Notification Service (SNS) - Publish/Subscribe Messaging

Amazon SNS is a **fully managed messaging service** that enables the delivery of notifications to various subscribers.

Subscribers Can Include:

- Email
- SMS
- AWS Lambda
- Amazon SQS

Summary & Key Takeaways

- AWS provides a variety of **compute services** beyond EC2.
- **ECS and EKS** help manage containerized applications.
- **Fargate** offers a **serverless container solution**.
- **Lambda and Step Functions** provide **serverless computing options**.
- **SQS and SNS** enable **asynchronous communication** between services.

By leveraging these AWS compute services, you can build scalable, event-driven, and cost-efficient applications.

🚀 **Congratulations! You've explored advanced AWS compute services!** 🎉

AI TUTOR ACTIONS

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Complete unit

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